$f : \mathbb{R}^n \longrightarrow \mathbb{R}$

$F : \mathbb{R}^n \longrightarrow \mathbb{R}^n$

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \ldots\ldots \frac{\partial f}{\partial x_n}(x) \right)$$ gradient

$$\nabla \cdot F(x) = \frac{\partial F_1}{\partial x_1}(x) + \ldots + \frac{\partial F_n}{\partial x_n}(x)$$ divergence

$$\nabla \times F(x) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{F_2}{\partial x} - \frac{F_1}{\partial y} \right)$$

$n = 3$

$$\Delta f = \frac{\partial^2 f}{\partial x_1^2} + \ldots + \frac{\partial^2 f}{\partial x_n^2}$$

$n = 3$

$\nabla \times \nabla f = 0$

$$\nabla \times \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2}{\partial z \partial y}, \dots, \dots \right) = \vec{0}$$

$$\nabla \cdot (\nabla \times F) = 0$$

$$\nabla \cdot (-y_z + z_y, -z_x + x_z, -x_y + y_x) =$$

$$\cancel{z_{yx}} - \cancel{y_{zx}} + \cancel{x_{zy}} - \cancel{z_{xy}} + \cancel{y_{xz}} - \cancel{x_{yz}} = 0$$

cotomological complex, ke imaš duc

zapoledn: pušči je 0

$$\mathbb{R} \xrightarrow{\nabla} \mathbb{R}^3 \xrightarrow{\nabla \times} \mathbb{R}^3 \xrightarrow{\nabla \cdot} \mathbb{R}$$

$K = [0,1] \times [0,1] \subseteq \mathbb{R}^2$

$$\int_K \nabla \cdot \vec{F}\, dx\, dy = \int_{\partial K} \vec{F}\, d\vec{n}$$

normale

$$\int_0^1 \int_0^1 \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} \right) dx\, dy = \int_0^1 dy \int_0^1 \frac{\partial F_1(x,y)}{\partial x} dx + \int_0^1 dx \int_0^1 \frac{\partial F_2(x,y)}{\partial y} dy$$

$$= \int_0^1 F_1(x,y)\Big|_0^1 dy + \int_0^1 F_2(x,y)\Big|_0^1 dx =$$

$$= \int_0^1 \left(F_1(1,y) - F(0,y) \right) dy + \int_0^1 \left(F_2(x,1) - F_2(x,0) \right) dx$$

$$\int_{\partial K} \vec{F}\, d\vec{n} = \int_0^1 \left(F_1(x,0), F_2(x,0) \right) \begin{pmatrix} dx \\ 0, -1 \end{pmatrix} dy + \dots$$

$$= \int_0^1 -F_2(x,0)\, dx + \dots = \int_0^1 \left(F_1(1,y) - F_1(0,y) \right) dy + \int_0^1 \left(F_2(x,1) - F_2(x,0) \right) dx$$

$u_{xy} = x^2 y \qquad u(x,0) = x \qquad u(0,y) = y^2$

$$u_x = \frac{1}{2}x^2y^2 + C(x)$$

$$u = \frac{1}{6}x^3y^2 + C(x) + D(y)$$

$$u(x,0) = C(x) + D(0) = x$$

$$u(0,y) = C(0) + D(y) = y^2$$

$$C(x) + D(y) = x + y^2$$

$C(0)$ in $D(0)$ inhlo dano za 0

$$u(x,y) = \frac{1}{6}x^3y^2 + x + y^2$$

$$u_{xy} + u_x = y e^{-y}$$

$$u(0,y) = 0$$

$$u(x,x) = 0$$

$$u_y(0,y) = 0$$

$$\frac{u(x,x)}{\partial y} = 0$$

integriramo po x

$$u_y + u = xye^{-y} + C(y)$$

v $(0,y) \rightsquigarrow$

$$0 + 0 = 0 + C(y) = 0$$

$$u_y + u = xye^{-y}$$

linearna diferencialna enačba

homogen del

$$u + u_y = 0$$

$$u = -\frac{\partial u}{\partial y} \quad dy = -\frac{du}{u} \rightsquigarrow y = -\ln u + C(x)$$

$$u = Ce^{-y}$$

partikularna

$$C'(y)e^{-y} = xye^{-y}$$

$$C'(y) = xy$$

$$C(x,y) = \frac{1}{2}xy^2$$

$$u = \frac{1}{2}xy^2e^{-y} + C(x)e^{-y}$$

$$u(0,y) = C(0)e^{-y} = 0$$

$$u(x,x) = \frac{1}{2}x^3e^{-x} + C(x)e^{-x} = 0$$

$$\Rightarrow C(x) = -\frac{1}{2}x^3$$

$$u(x,y) = \frac{1}{2}x\,e^{-x}(y^2 - x^2)$$

c)

$u_x + u_y = 0 \qquad u(x,0) = x^2$

$$u(x,y) = \sum_{i,j} a_{ij} x^i y^j$$

$$u(x,0) = \sum_i a_{i0} x^i = x^2 \Rightarrow a_{20} = 1$$

$a_{i0} = 0$ za $i \neq 2$

$$u_x = \sum_{i,j} a_{ij}\, i\, x^{i-1} y^j = a_{10} + \sum_{i,j\geq 0} a_{i+1\,j}\,(i+1) x^i y^j$$

$$u_y = \sum_{i,j} a_{ij}\, j\, x^j y^{j-1} = a_{01} + \sum_{i,j\geq 0} a_{i\,j+1}\,(j+1)\, y^j x^i$$

$$u_x + u_y = a_{01} + a_{10} + \sum_{i,j\geq 1} (a_{i+1,j}(i+1) + a_{i,j+1}(j+1)) x^i y^j = 0$$

$$\Rightarrow (a_{i+1})(i+1) + a_{i,j+1}(j+1) = 0 \text{ za } \forall\, xy$$

$a_{01} + a_{10} = 0 \Rightarrow a_{01} = -a_{10}$

$\Rightarrow a_{0j} = 0$ razen $a_{02} = -2$

$a_{2,1} = a_{1,2}$

$a_{11} = -a_{0,2}$

$a_{11} = -a_{2,0}$

$a_{1,0} + a_{01} = 0,\ a_{1,1} + a_{02} = 0 \Rightarrow a_{11} = 2$

j \ i	0	1	2	3		
0	a_{00}	0	1	0	0	...
1	0	-2	0	0		
2	1	0	0			
3	0	0				
	0					

$a_{3,0} = -a_{2,1}$

$a_{11} = -a_{02}$

$$u(x,y) = a_{00} + a_{2,0} x^2 + a_{11} xy + a_{02} y^2$$

$$= a_{00} + x^2 - xy + y^2$$

$$u(x,0) = a_{00} + x^2 \Rightarrow a_{00} = 0$$

$\Rightarrow u(x,y) = x^2 - xy + y^2 \qquad a_{2,0} = a_{11}$

Hmmm nekaj je narobe

$u(x,y) = x^2 - 2xy + y^2$

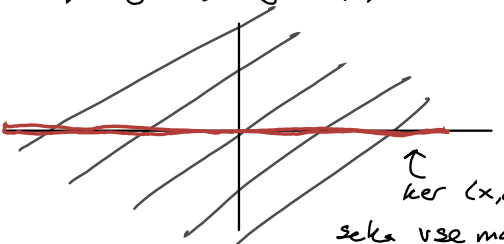
Pozabila sem množiti z $i+1$ in $j+1$ pri odvajanju

3. c)

$u_x + u_y = 0 \qquad u(x,0) = x^2 \quad \rightsquigarrow u(x,y) = (x-y)^2$

Naj bo $u(x,y) = g(x-y)$

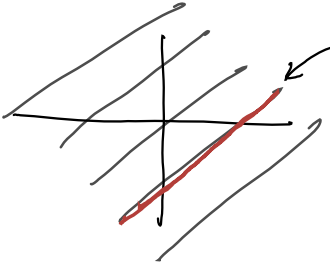
$\Rightarrow u_x + u_y = g'(x-y) - g'(x-y) = 0$



ker $(x,0)$ seka vse možnosti, je to validn način

Slab primer

$u(x,x) = x^2$



če imamo tak robni pogoj ne moremo dobiti rešitve

Ali neobstaja, ali pa jako je veliko

$$u_{xy} + y u_y + u = 1 \qquad u_x(x,0) = 1$$
$$u(x,x) = e^{-x^2}$$

$$y u_y + u = (yu)_y$$

$$u_{xy} + (yu)_y = 1 \rightsquigarrow u_x + yu = y + C(x)$$

$$V(x,0): \quad \overset{0}{\overset{\shortparallel}{u_x(x,0)}} + 0 \cdot 1 = 0 + C(x) \Rightarrow C(x) = 0$$

$$u_x + yu = y$$

homogeno:

$$u_x = -yu$$
$$\frac{\partial u}{u} = -y\,dx$$
$$u = D(y) e^{-yx}$$

partikularna:

$$D_x(x,y) e^{-yx} = y$$
$$\rightsquigarrow D(x,y) = \int y e^{+yx}\,dx = e^{+yx} + C(y)$$

$$u = 1 + C(y) e^{-yx}$$
$$u(x,x) = 1 + C(x) e^{-x^2} = e^{-x^2} \Rightarrow C(x) = \frac{e^{-x^2} - 1}{e^{-x^2}} = 1 - e^{x^2}$$

$$u(x,y) = 1 + (1 - e^{-y^2}) e^{-yx}$$

$$u_{xy} + 2u_y = e^{-2x} \qquad \begin{cases} u(x,0) = e^{-x} \\ u_x(x,0) = -e^{-x} \end{cases} \qquad u(0,y) = 1$$

---

integriamo per y

$$\rightsquigarrow \quad u_x + 2u = ye^{-2x} + C(x)$$

$$v(x,0): \quad -e^{-x} + 2e^{-x} = C(x) = e^{-x}$$

$$u_x + 2u = ye^{-2x} + e^{-x}$$

hom:

$$\frac{\partial u}{u} = -2dx \Rightarrow u = e^{-2x} C(y)$$

part: $\quad C_x(x,y)e^{-2x} = ye^{-2x} + e^{-x}$

$$C(x,y) = \int (y + e^{x})dx = yx + e^{x} + D(y)$$

$$u = yxe^{-2x} + e^{-x} + e^{-2x} D(y)$$

$$v(0,y): \quad 0 + 1 + D(y) = 1 \Rightarrow D(y) = 0$$

$$\Rightarrow u = yxe^{-2x} + e^{-x}$$

KLE

$$a(x,y,u)\,u_x + b(x,y,u)\,u_y = c(x,y,u)$$

$S = \{ (x,y,z) \in \mathbb{R}^3 ;\quad z = u(x,y) \quad u \text{ rešitev} \}$

je integralska krivulja

Presek $S_1 \cap S_2$ je karakteristična krivulja

Karakteristična krivulja γ: $\dot{\gamma}(t) = (a(\gamma(t)), b(\gamma(t)), c(\gamma(t)))$

Izrek:

- S_1, S_2 integralski ploskvi za isto KLE. $p \in S_1 \cap S_2 \Rightarrow$ karakteristična krivulja, ki vsebuje p leži cela v $S_1 \cap S_2$
- $S_1 \cap S_2$. če se ne sekata tangentno, je karakteristična krivulja

Dokaz 2)

$p \in C = S_1 \cap S_2$

$\Pi_1 = T_p S_1 \qquad \Pi_2 = T_p S_2$

$\dim(\Pi_1 \cap \Pi_2) = 1$, ker se ne sekata S_1 in S_2 tangentno.

$\Pi_1 \cap \Pi_2$ tangentna na C

$$a(p), b(p), c(p) \in \Pi_1 \cap \Pi_2 = T_p C$$

$$T_p C = \{\lambda (a(p), b(p), c(p)) + p \quad ; \lambda \in \mathbb{R}\}$$

C je tangentna na karakteristično smer

$\Rightarrow C$ je karak. krivulja

1

$$x u_x + y u_y = 0 \qquad (x, y, 0) = (a, b, c)$$

$$\dot{x} = x \rightsquigarrow x_0(s) e^t$$

$$\dot{y} = y \rightsquigarrow y_0(s) e^t$$

$$\dot{z} = 0 \rightsquigarrow z = z_0(s)$$

$$\varphi(t) = u(tx, ty)$$

$$\dot{\varphi}(t) = u_x \cdot x + u_y \cdot y = 0$$

$$\Rightarrow \varphi(t) = C(s) = C \qquad \varphi(0) = u(0,0)$$

$$\Rightarrow \varphi(t) = u(0,0)$$

$$\varphi(1) = u(x,y) = C = u(0,0)$$

$$\Rightarrow u(x,y) = u(0,0)$$

② $u : \bar{D} \to \mathbb{R}$

$a(x,y)\, u_x + b(x,y)\, u_y = -u$

$x\, a(x,y) + y\, b(x,y) > 0$ za $x,y \in \partial D$

min, max $D \Rightarrow u \equiv 0$

max $\in D \Rightarrow u_x = u_y = 0$ v (x_0, y_0)

$\Rightarrow -u(x_0, y_0) = 0$

min $\in D \Rightarrow -u(x_1, y_1) = 0 \Rightarrow$ $\Rightarrow u \equiv 0$

b) $(x,y,\lambda) \longmapsto u(x,y) + \lambda(x^2+y^2-1)$

$u_x(x,y) + 2\lambda x = 0$

$u_y + 2\lambda y = 0$

$x^2+y^2-1 = 0$

c) $t \xmapsto{f} u(tx_0, ty_0)$ x_0, y_0 maks na robu

$\dot{f} = u_x x_0 + u_y y_0 \geq 0$ v okolici $t=1$

$t=1: u_x(x_0,y_0)\, x_0 + u_y(x_0,y_0)\, y_0 = -2\lambda(x_0 + y_0)$

$\Rightarrow \lambda \leq 0$

Denimo da $u(x_0,y_0) > 0$

$a(x,y)(-2\lambda x_0) + b(x,y)(-2\lambda y_0) \overset{=-u}{<} 0$

$(-2\lambda)\underbrace{(a(x,y)x_0 + b(x,y)y_0)}_{>0} < 0$

$\Rightarrow -2\lambda < 0 \Rightarrow \lambda > 0$ $*$

Sledi $u(x_0,y_0) < 0$

kjer koli je minimum ne v kotu $\geq 0 \Rightarrow$

$u \equiv 0$

Metoda karakteristik za enačbe 1. reda

enačba je

- $a(x, u(x)) \cdot Du = b(x, u(x))$
- $u(x) = g(x)$ .... začetni pogoj
  na krivulji Γ $\quad g : \Gamma \to \mathbb{R}$

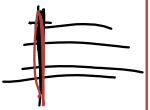
① Rešimo sistem za vsaj po s

$$\dot{z}(y,s) = b(x(y,s), z(y,s))$$

odvod po s

$$\dot{x}(y,s) = a(x(y,s), z(y,s))$$

$$x(y,0) = y \quad ; \quad z(y,0) = g(y)$$



② Najdemo $s(x)$, $y(x)$, potem $u(x) = z(y(x), s(x))$

(1) $u(0,x) = g(x)$

$u_t + a \cdot \nabla u = 0$

$\dot{z}(y,s) = 0 \quad \rightsquigarrow \quad z = c(y)$

$\dot{t}(y,s) = 1 \quad \rightsquigarrow \quad t = 1s + c(y)$

$\dot{x}(y,s) = a \quad \rightsquigarrow \quad x = as + c(y)$

$(t(y,0), x(y,0)) = (0,y)$

$z(y,0) = g(y)$

$z = c(y) = g(y)$

$t(y,0) = c(y) = 0 \rightsquigarrow t(y,s) = s$

$x(y,0) = c(y) = y \rightsquigarrow x(y,s) = as + y$

$y = x - at$

$u(x,y) = g(x - at)$

b)

$$u_t + a\partial u + u = f(x)$$

$$u_t + a\partial u = f(x) - u$$

$$\dot{z}(y,s) = f(x(y,s)) - z(y,s)$$

$$\dot{t}(y,s) = 1 \rightsquigarrow t = s + C(y)$$

$$\dot{x}(y,s) = a \rightsquigarrow x = s + D(y)$$

$$x(y,0) = g(y) \rightsquigarrow x = s + g(y)$$

$$t(y,0) = 0 \rightsquigarrow t = s$$

$$\rightsquigarrow \dot{z}(y,s) = f(s + g(y)) - z$$

hom: $\int \frac{dz}{z} = -\int ds \rightsquigarrow z = C(y)e^{-s}$

par: $\dot{C}(y,s)e^{-s} = f(s + g(y))$

$$\rightsquigarrow C = \int f(s + g(y))\, e^{s})\, ds$$

$$z(y,s) = \int f(t + g(y))\, e^{t}\, dt\; e^{-s}$$

$$= \int f(t)\, e^{t}\, dt\; e^{-g(y)} e^{-s}$$

$$u(x,y) = \left(\int f(t)\, e^{t}\, dt\right) e^{-x}$$

c) $u_t + \nabla F(u) = 0$

$$\nabla F(u(x_1, \dots, x_n)) = \left(F'(u)\frac{\partial u}{\partial x_1}, \dots, F'(u)\frac{\partial u}{\partial x_n} \right) =$$
$$= F'(u) \cdot \nabla u$$

$\dot{z} = 0 \quad \rightsquigarrow \quad z = c(y) = g(y)$

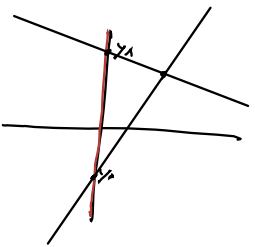
$\dot{x} = F'(z) \rightsquigarrow \dot{x} = F'(c) \rightsquigarrow x = F'(g(y))s + y$

$\dot{t} = 1 \quad \rightsquigarrow \quad t = s + D(y) \rightsquigarrow t = s$

$t(y,0) = 0$

$x(y,0) = y$

$z(y,0) = g(y)$



$x(y,s) = F'(g(y))\, s + y$

kej ko se seceta

$F'(g(y_0))s + y_0 = F'(g(y_1))s + y_0$

če se seceta mora biti $g(y_0) = g(y_1)$

torej če se seceta, se ne more zgoditi
analitična rešitev obstaja brezhibno

a) $uu_x + u_y = 1 \qquad u(x,x) = \frac{x}{2}$

$a = u \qquad \dot{x} = z \rightsquigarrow x = \frac{1}{2}t^2 + \frac{s}{2}t + C(s) \rightsquigarrow$

$b = 1 \qquad \dot{y} = 1 \rightsquigarrow t + C(s) \rightsquigarrow y = t + s$

$c = 1 \qquad \dot{z} = 1 \rightsquigarrow t + C(s) \rightsquigarrow z = t + \frac{s}{2}$

$x(s,0) = s$

$y(s,0) = s$

$z(s,0) = \frac{s}{2}$

$x = \frac{1}{2}t^2 + \frac{1}{2}st + s \rightsquigarrow s = -$

$t = y - s \rightsquigarrow s = y - t$

$u(x,y) = \frac{s}{2} + t = \frac{s}{2} + y - s = y - \frac{s}{2}$

$x = \frac{1}{2}t^2 + \frac{1}{2}yt - t^2 + y - t = -\frac{1}{2}t^2 + \frac{1}{2}yt + y - t$

$\vdots \qquad s = \frac{3x - y^2}{2 - y}$

čo $x \neq 2$ je vrcholu

a z. vrch. $y \neq 2$

b) $-\sin(y)\, u_x + x u_y = 0 \qquad u(x,0) = g(x) \quad za\ x>0$

$$\left.\begin{aligned} \dot{x} &= -\sin(y) \\ \dot{y} &= x \end{aligned}\right\} \qquad \ddot{y} = \dot{x} = -\sin(y)$$

$$\frac{d^2}{dt^2}\sin(y) = \frac{d}{dt}\cos y \cdot \dot{y} = \ddot{y}\cos y + \dot{y}(-\sin y)\dot{y}$$

$$\dot{z} = 0 \rightsquigarrow z = g(s)$$

$$x(s,0) = x$$

$$y(s,0) = 0$$

$$z(s,0) = g(s)$$

1. energija $E_1(x,y,z) = z$

$$\ddot{y} = \dot{x} = -\sin(y) \qquad x\dot{x} = \cancel{x}\ddot{y} = \dot{y}\ddot{y} = -x\sin(y) = -\dot{y}\sin y$$

2. energija

$$x\dot{x} = (\cos y)^{\cdot}$$

$$\int x\,dx \int (\cos y)^{\cdot}\,dt = \cos y + C$$

$$\frac{1}{2}x^2 = \cos y + c$$

$$E_2 = \frac{1}{2}x^2 - \cos y$$

Iščemo H:

$$E_1(x(0), y(0), z(0)) = g(s)$$

$$E_2(x(s,0), y(s,0), z(s,0)) = \frac{1}{2}s^2 - 1$$

$$H(a,b) = a - g(\sqrt{2b+2})$$

$$H(E_1, E_2) = g(s) - g(\sqrt{s^2}) = 0$$

$$H(E_1(x,y,z), E_2(x,y,z)) = 0 \quad za\ vse\ t,s$$

$$\overset{\shortparallel}{z}$$

$$z - g(\sqrt{x^2 - 2\cos y + 2}) = 0$$

$$\Rightarrow u(x,y) = g(\sqrt{x^2 - 2\cos y + 2})$$

c) $x(y^2+u)u_x - y(x^2+u)u_y = (x^2-y^2)u$

$u(x,x) = c \in \mathbb{R}$ za $x \neq 0$

$\dot{x} = x(y^2+z)$

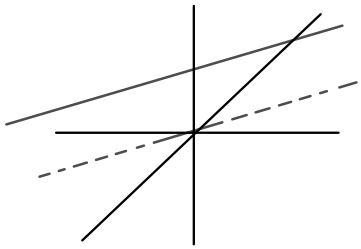
$\dot{y} = -y(x^2+z)$

$\dot{z} = z(x^2-y^2) = z(x-y)(x+y)$

$x(s,0) = s$

$y(s,0) = s$

$z(s,0) = c$



$\dot{x}(s,0) = s(s^2+c)$

$\dot{x}x + \dot{y}y = z(x^2-y^2) = \dot{z}$

$\frac{x^2}{2} + \frac{y^2}{2} = z + D(s)$

$E_1(x,y,z) = \frac{x^2}{2} + \frac{y^2}{2} - z$

$\frac{\dot{x}}{x} + \frac{\dot{y}}{y} + \frac{\dot{z}}{z} = y^2 - x^2 + x^2 - y^2 = 0$

$\ln x + \ln y + \ln z = c$

$xyz = F(s) \qquad E_2(x,y,z) = xyz$

$E_1(s,0) = s^2 - c = a$

$E_2(s,0) = cs^2 = b \qquad ca - b + c^2$

$H(a,b) = ca - b + c^2$

$H(E_1,E_2) =$

$c\left(\frac{x^2}{2} + \frac{y^2}{2} - z\right) - xyz + c^2 = 0$

izrazimo z

$$z = u(x,y) = \frac{c^2 + \frac{c}{2}(x^2+y^2)}{c + xy}$$

Energija

Pri uporabi karakteristik v n-dimenzionalnem sistemu, lahko iščemo n "energij". To so funkcije $E_j(x(t))$ za katere velja, da so

$$\frac{d}{dt} E_j(x(t)) = 0$$

Nato predpišemo $H(E_1, \ldots, E_n) = 0$ (oz poljubno konstanto) in z uporabo začetnih pogojev določimo H, ki zadošča $H(E_1(x(0)), \ldots) = 0$

$u_x + x u_y = u \qquad u(1,y) = h(y) \qquad h \in \mathcal{C}(\mathbb{R})$

$\dot{x} = 1 \qquad x(s,0) = 1$

$\dot{y} = x \qquad y(s,0) = s$

$\dot{z} = z \qquad z(s,0) = h(y)$

$x = t + 1$

$y = \frac{1}{2}t^2 + t + s$

$z = h(y)\, e^{x-1}$

$\dot{z} = z$

$\frac{dz}{z} = dt$

$z = ce^t = h(y)e^{x-1}$

$t = x - 1$

$s = y - \frac{1}{2}(x-1)^2 - (x-1)$

$u(x,y) = h\left(y - \frac{1}{2}(x-1)^2 - (x-1)\right)e^{x-1}$

$$x^2 u_x + y^2 u_y = (x+y)u \qquad u(x,1)=1$$
$$x(s,0)=s$$

$$\dot{x} = x^2 \rightsquigarrow -\frac{1}{x} = t + c \rightsquigarrow x = \frac{-1}{t+c} \rightsquigarrow x = \frac{s}{1-ts}$$

$$\dot{y} = y^2 \rightsquigarrow y = \dots = \frac{1}{1-t}$$

$$\dot{z} = (x+y)z$$

$$\frac{\dot{z}}{z} = \frac{1}{1-t} + \frac{s}{1-ts}$$

nekaj pride se zaradi nagaj

$$\ln z = -\ln(1-t) - \ln(1-ts) +$$

$$z = \cancel{\frac{1-ts}{1-t}} \quad \frac{1}{(1-t)(1-ts)}$$

$$z = y \cdot \frac{x}{s}$$

$$s = x(1-ts)$$

$$s(1+tx) = x \rightsquigarrow s = \frac{x}{1+tx}$$

$$(1-t)y = 1$$

$$t = -\frac{1}{y} + 1 = 1 - \frac{1}{y} = \frac{y-1}{y}$$

$$s = \frac{x}{1 - \frac{(y-1)x}{y}} = \frac{xy}{y+x-xy}$$

$$z = y+x-xy \rightsquigarrow u(x,y) = y+x-xy$$

$u(x,0) = 1$

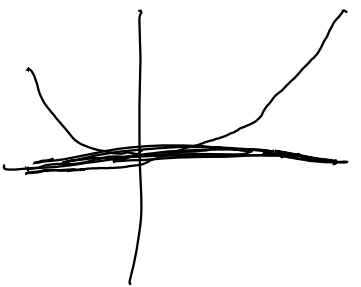
$$x = \frac{1}{t+c}$$
$$y = \frac{1}{t+c}$$

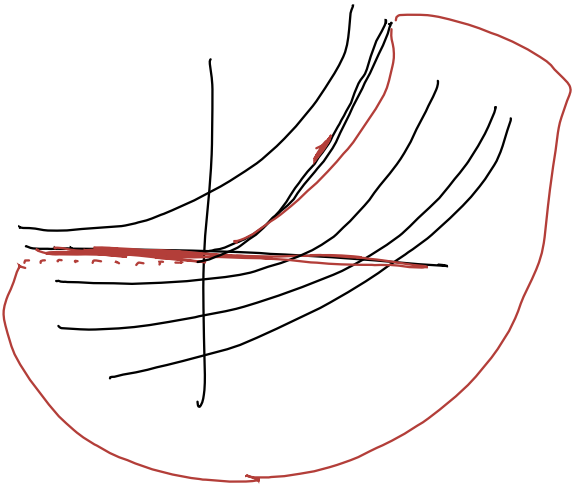
$$x(s,0) = s$$
$$y(s,0) = 0$$
$$z(s,0) = 1$$

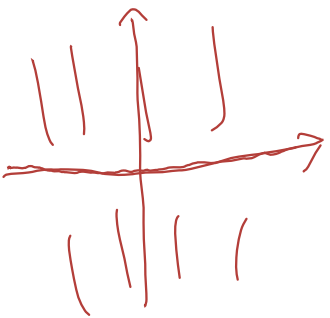
$$\frac{1}{c} = s \rightsquigarrow c = \frac{1}{s} \rightsquigarrow x = \frac{s}{1-ts}$$
$$\frac{1}{c} = 0 \rightsquigarrow c = \infty \qquad y \equiv 0$$

$$\dot{z} = \frac{s}{1-ts} z$$
$$\ln z = -\ln(1-ts) + C \rightsquigarrow$$

$$z = \frac{E(s)}{1-ts} = \frac{1}{1-ts}$$ ne marelamo izrazi z x in y $\rightsquigarrow$ ne dobimo enačbe







ogrinjača družine ravnin skozi točko

Naj bo $\Pi_\lambda = \{(x,y,z) \mid z = \psi(x,y,\lambda) = a(\lambda)x + b(\lambda)y\}$
družina ravnin skozi izhodišče
$a, b \in \mathcal{C}(I)$ $I \subseteq \mathbb{R}$ interval, da je
$\vec{n}(\lambda) = (a(\lambda), b(\lambda), -1)$ injektivna
in $w(a', b') = \begin{vmatrix} a' & b' \\ a'' & b'' \end{vmatrix} \neq 0$

Opazimo: $z = -1$ komponenta je normalizacija, ki da $\Pi_{\lambda_1} = \Pi_{\lambda_2} \Leftrightarrow \lambda_1 = \lambda_2$

Cilj: Najdemo $\varphi(x,y)$, ki definira $\sigma = \{(x,y,\varphi(x,y))\}$
da velja
- $\sigma \subseteq \bigcup_\lambda \Pi_\lambda$
- $P_0 = (x_0, y_0, z_0) \in \sigma \cap \Pi_\lambda$ za nek λ

$\Longrightarrow$ normalni vektor za σ v točki p_0 je sorazmeren z $\vec{n}(\lambda)$

Opazka: $\sigma \equiv \Pi_\lambda$ za fiksen λ je trivialna rešitev, ki jo ignoriramo

$(\bigcup \Pi_\mu)$

Ker je $\vec{n}(\lambda)$ injektivna, če $p_0 \in \sigma \cap \Pi_\lambda$ in $p_0 \in \sigma \cap \Pi_\mu$ potem $\lambda = \mu$

Consider the map $P: \mathbb{R}^3 \to \mathbb{R}^2$
če $(x,y) = P(x,y,z)$ za nek $(x,y,z) \in \sigma$
potem $p \in \Pi_\lambda$ za nek unikaten $\lambda \in \Lambda(x,y)$

$\Lambda : P(\sigma) \to \mathbb{R}$
$(x,y) \mapsto \Lambda(x,y)$

Ker $\sigma = \{z = \varphi(x,y)\} \subseteq \bigcup_\lambda \Pi_\lambda$ imamo
$\varphi(x,y) = a(\Lambda(x,y))x + b(\Lambda(x,y))y = \psi(x,y,\Lambda(x,y))$

normala za σ je paralelna $\vec{n}(\lambda)$, torej
parametrizacija σ je $u(x,y) = (x,y,\varphi(x,y))$
normala je podana z $u_x \times u_y$
računamo:
$u_x = (1, 0, \varphi_x(x,y)) = \psi_x(x,y,(\lambda(x,y)) + \psi_\lambda(x,y,(\lambda(x,y))\Lambda_x(x,y)$
$u_y = (0, 1, \varphi_y(x,y)) = \psi_y(x,y,\Lambda(x,y)) + \psi_\lambda(x,y,\Lambda(x,y))\Lambda_y(x,y)$

$-u_x \times u_y = (\psi_x + \psi_\lambda \Lambda_x, \psi_y + \psi_\lambda \Lambda_y, -1)$
$= (\psi_x, \psi_y, -1) + \psi_\lambda(\Lambda_x, \Lambda_y, 0) =$

Spomnimo se $\psi = a(x)x + b(x)y$

$= \underbrace{(a(\Lambda(x,y)), b(\Lambda(x,y)), -1)}_{\vec{n}(\Lambda(x,y))} + \psi_\lambda(\Lambda_x, \Lambda_y, 0)$

Želimo dalje $= 0$

1. $(\Lambda_x, \Lambda_y, 0) = 0 \Rightarrow \nabla\Lambda = 0 \Rightarrow \Lambda$ konstantna
$\Rightarrow \psi(x,y) = a(\lambda)x + b(\lambda)y \Rightarrow \sigma \subseteq \Pi_\lambda$ trivialna rešitev

2. $\psi_\lambda(x,y,\lambda(x,y)) = 0 \quad \forall x,y \in P(\sigma)$

vsaen ravnina je, ravnina } pri filiju

a)

$(u_x u_y)^2 = u \qquad u(x,0) = x$

$(pq)^2 = z$

$F(x,y,z,p,q) = (pq)^2 - z$

$\dot{x} = \cancel{pq} = 2(pq)q$

$\dot{y} = 2(pq)p$

$\dot{z} = 2p^2q^2 + 2p^2q^2 = 4p^2q^2$

$\dot{p} = -(-p) = p$

$\dot{q} = q$

$p = C(s)e^t$

$q = De^t$

$z = 4\dot{z} \rightsquigarrow z = E(s)e^{4t}$

$\dot{z} = 4e^{4t}CD$

$z = e^{4t}C^2D^2 = (e^{4t}-1)C(s)^2D(s)^2 + E(s)$

$\dot{x} = 2C^2De^{3t} \rightsquigarrow x = \frac{2}{3}C^2D(e^{3t}-1) + F(s)$

$\dot{y} = \frac{2}{3}CD^2e^{3t} \rightsquigarrow x = \frac{2}{3}CD^2(e^{3t}-1) + G(s)$

$u(x,0) = x$

$x(s,0) = s = F(s)$

$y(s,0) = 0 = G(s)$

$z(s,0) = s = E(s)$

$p(s,0) = u_x(x(s,0), y(s,0)) = 1 \rightsquigarrow C(s) = 1$

$q(s,0) = \pm\sqrt{s}$

$(u_x u_y)^2(x,0) = u(x,0) = s$

$(1 \cdot u_y(x,0))^2 = x$

$q_0 = u_y(x,0) = \pm\sqrt{s}$

$u(x,0) = \pm\sqrt{x}\,y + c(x)$

$z = (e^{4t}-1)s + s$

$x = \frac{2}{3}(e^{3t}-1)s + s \qquad s = \dfrac{x}{1+\frac{2}{3}(e^{3t}-1)}$

$y = \frac{2}{3}(e^{3t}-1)\sqrt{s}$

$p = e^t \qquad e^{3t} = \frac{3}{2}\frac{y}{\sqrt{s}} + 1$

$q = e^t\sqrt{s}$

$s = \dfrac{x}{1+\frac{y}{\sqrt{s}}}$

$s + \sqrt{s}\,y = x$

$s = \left(\dfrac{y}{\frac{2}{3}(e^{3t}-1)}\right)^2$

$x = \dfrac{y^2}{\underbrace{\frac{2}{3}(e^{3t}-1)}_{v}} + \left(\dfrac{y}{\frac{2}{3}(e^{3t}-1)}\right)^2$

$s = \left(\dfrac{x^2}{1 \pm \sqrt{y^2-4x}}\right)^2$

$v^2x - vy^2 - y^2 = 0$

$D = \sqrt{y^4 - 4xy^2} = y\sqrt{y^2-4x}$

$\frac{2}{3}(e^{3t}-1) = \dfrac{y^2 \pm y\sqrt{y^2-4x}}{x^2}$

$u = \frac{3}{2}\left(\dfrac{y^2 \pm y\sqrt{y^2-4x}}{x^2}\right)\left(\dfrac{x^2}{1\pm\sqrt{y^2-4x}}\right)^2 + \left(\dfrac{x^2}{1\pm\sqrt{y^2-4x}}\right)^2$

b) $u_x^2 + u_y = yu$ $\qquad u(x,0) = x$

$F(x,y,z,p,q) = p^2 + q - yz$

$$\dot{x} = 2p \qquad x(s,0) = s$$
$$\dot{y} = 1 \qquad y(s,0) = 0$$
$$\dot{z} = p\cdot(2p) + q\cdot 1 \qquad z(s,0) = s$$
$$\dot{p} = -0 + py = py \qquad p(s,0) = 1$$
$$\dot{q} = z + qy \qquad q(s,0) = -1$$

$y = t + c(s) = t$

$dp = p\,dt \rightsquigarrow \ln p = \frac{1}{2}t^2 dt + c \rightsquigarrow p = p_0 e^{\frac{1}{2}t^2} = e^{\frac{1}{2}t^2}$

$dx = 2e^{\frac{1}{2}t^2} dt \rightsquigarrow x =$

$$2\int e^{\frac{1}{2}t^2} dt = \int \sqrt{2}\, u^{-\frac{1}{2}} e^u \, du$$

$u = \frac{1}{2}t^2$

$du = t\,dt = \sqrt{2u}\,dt$

$\dot{z} = 2e^{t^2} + q$

$\dot{q} = z + qt$

$\ddot{z} = \ddot{q} + q + t\dot{q} = 2e^{t^2} + q$

$\ddot{q} + t\dot{q} = 2e^{t^2}$

hom: $\frac{d\dot{q}}{\dot{q}} = -t\,dt \rightsquigarrow \dot{q} = A(s)e^{-t}$

pt: $\qquad A' e^{-t} = 2e^{t^2}$

$A' = 2e^{t^2+t}$

b) $u_x^2 + u_y = yu$ $\qquad u(x,0) = x$

$F(x,y,z,p,q) = p^2 + q - yz$

$$\dot{x} = 2p \qquad x(s,0) = s$$
$$\dot{y} = 1 \qquad y(s,0) = 0$$
$$\dot{z} = p\cdot(2p) + q\cdot 1 \qquad z(s,0) = s$$
$$\dot{p} = -0 + py = py \qquad p(s,0) = 1$$
$$\dot{q} = z + qy \qquad q(s,0) = -1$$

Linearne PDE 2. reda v $\mathbb{R}^2$

$$\underbrace{a(x,y)u_{xx} + 2b\,u(x,y)u_{xy} + c u_{yy}} + d u_x + e u_y + f u + g = 0$$

$$\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right)\begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{pmatrix} u$$

$$\Delta = b^2 - ac = -\det\begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

		kanonična oblika
$\Delta > 0$	hiperbolični tip	u_{xy} + niži red $= 0$
		oziroma $u_{xx} - u_{yy}$ + niž. $= 0$
$\Delta = 0$	parabolični tip	u_{xx} + niž $= 0$
$\Delta < 0$	eliptični tip	$u_{xx} + u_{xy}$ + niž. $= 0$

Hiperbolični tip:

$$\frac{\partial y}{\partial x} = \frac{b \pm \sqrt{\Delta}}{a} \rightsquigarrow y = f_\pm(x, C)$$

Novi koordinati $\xi(x,y)$, $\eta(x,y)$

velja $y = f_+(x; \xi(x,y))$, $y = f_-(x, \eta(x,y))$

Primer: $\frac{dy}{dx} = 2 \pm 1 \rightsquigarrow y = \begin{cases} 3x + C = f_+ \\ x + C = f_- \end{cases}$

$$\xi(x,y) = y - 3x$$

$$\eta(x,y) = y - x$$

- če $a = 0 \rightsquigarrow \frac{dx}{dy} = \frac{b \pm \sqrt{D}}{c}$
- če $a = 0$ in $c = 0 \rightsquigarrow$ enačba je že v kanonični obliki

Parabolični tip $\Delta = 0$

$$\frac{dy}{dx} = \frac{b}{a} \quad \left(= \frac{b \pm \sqrt{\Delta}}{a}\right) \rightsquigarrow y = f(x, C)$$

$$y = f(x, \xi(x,y))$$

$\eta(x,y)$ je poljubna (ampak še koordinati)

Eliptični tip $\Delta < 0$

$$\frac{dy}{dx} = \frac{b \pm i\sqrt{-\Delta}}{a} = \frac{b \pm \sqrt{\Delta}}{a} \rightsquigarrow y = f(x, C)$$

Iščemo $\Phi(x,y)$: $y = f(x, \Phi(x,y))$

$$\xi = \mathrm{Re}\,\Phi \qquad \eta = \mathrm{Im}\,\Phi$$

$$u_{xx} + 2u_{xz} + u_{yy} + u_{zz} = 0$$

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Diagonalizirano

$$(\lambda I - A) = \begin{bmatrix} \lambda-1 & 0 & -1 \\ 0 & \lambda-1 & 0 \\ -1 & 0 & \lambda-1 \end{bmatrix} =$$

$$= (\lambda-1)\left((\lambda-1)^2-1\right) = (\lambda-1)(\lambda^2-2\lambda) =$$
$$\lambda(\lambda-1)(\lambda-2)$$

$\lambda=0$: $e_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ $\qquad Q = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

$\lambda=1$: $e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$\lambda=2$: $e_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ $\qquad D = \begin{bmatrix} 0 & & \\ & 1 & \\ & & 2 \end{bmatrix}$

$$A = Q^T D Q$$

$\xi_1 = x - z$

$\xi_2 = y$ $\qquad \xi = Q\begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$\xi_3 = x + z$

$u = u(x(\xi), y(\xi), z(\xi)) = v(\xi)$

$x = \frac{1}{2}(\xi_1 + \xi_3)$

$u_x = (\xi_1)_x v_{\xi_1} + (\xi_2)_x (v_{\xi_2}) + (\xi_3)_x v_{\xi_3}$ $\qquad y = \xi_2$

$z = \frac{1}{2}(\xi_3 - \xi_1)$

...

$u_x = v_{\xi_1} + v_{\xi_3}$ $\qquad \partial_x = \partial_{\xi_1} + \partial_{\xi_3}$

$u_y = v_{\xi_2}$ $\qquad \partial_y = \partial_{\xi_2}$

$u_z = -v_{\xi_1} + v_{\xi_3}$ $\qquad \partial_z = \partial_{\xi_3} - \partial_{\xi_1}$

$u_{xx} = v_{\xi_1} + 2v_{\xi_1\xi_3} + v_{\xi_3}$

$u_{yy} = v_{\xi_2\xi_2}$

$u_{zz} = v_{\xi_1} \cancel{-2v_{\xi_1\xi_3}} + v_{\xi_3\xi_3}$

$u_{xz} = -v_{\xi_1\xi_1} + \ldots$

$$u_{xx} + 2u_{xz} + u_{yy} + u_{zz} = 0$$

$\Rightarrow$ $4v_{22} + v_{11} = 0$ $\qquad$ ni kanonična oblika

Moralo bi normirati Q

$\tilde{\xi}_2 = \frac{\xi_2}{2} \rightsquigarrow \tilde{v}_{22} = \frac{\xi_2}{2}$ $\qquad \tilde{v}(\xi_1, \tilde{\xi}_2, \xi_3)$

$= v(\xi_1, \frac{\xi_2}{2}, \xi_3$

$\tilde{v}_2 = v_2 \cdot \frac{1}{2}$

$\tilde{v}_{22} = v_{22} \cdot \frac{1}{4}$

$\Rightarrow$ $\tilde{v}_{22} + v_{11} = 0$

D'Alzimbertova formula na $[0,\infty) \times \mathbb{R}$

$$u(t,x) = \frac{1}{2}(f(x-t) + f(x+t)) + \frac{1}{2}\int_{x-t}^{x+t} g(\xi)\,d\xi + \frac{1}{2}\iint_{\Delta} F(\tau,\xi)\,d\tau\,d\xi$$

kjer je $\Delta(t,x) = \{(\tau,\xi) \in \mathbb{R}^2 \mid 0 \le \tau \le t,\ |\xi - x| \le t-\tau\}$

$$u_{tt} - u_{xx} = F(t,x)$$
$$u(0,x) = f(x)$$
$$u_t(0,x) = g(x)$$

$$u_{tt} - u_{xx} = 0$$
$$u(0,x) = u_t(0,x) = \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$$

$k, \omega > 0$

$$u_{tt} - u_{xx} = \sin(kx)\cos(\omega t)$$

$$u(0,x) = u_t(0,x) = 0$$

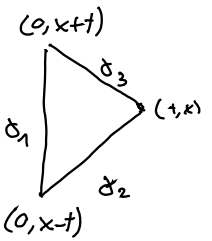
$$f(x) = g(x) = 0$$

$$F(x,t) = \sin(kx)\cos(\omega t)$$

$$u(t,x) = \frac{1}{2}\iint_{\Delta} F(\tau,\xi)\,d\tau\,d\xi$$

$$\Delta = \{(\tau,\xi)\in\mathbb{R}^2 \mid 0\le\tau\le t,\ |\tau - x| \le t-\tau\}$$

$$-\iint_D (Y_x - X_y)\,dx\,dy = \int_{\partial D} (X,Y)\,ds$$



$$X = -\frac{1}{\omega}\cos(kx)\cos(\omega t)$$

$$Y = 0$$

$$X_y = \sin(kx)\cos(\omega t)$$

$$\iint_{\Delta} \dots = -\int \left(-\frac{1}{\omega}\cos(kx)\cos(\omega t),\ 0\right) ds$$

$\gamma_1 : 0$

$\gamma_2 :\ \gamma_1(s) = (s,\ x-(t-s))\quad s\in(0,t)$

$\gamma_3 = (t-s,\ x+s)\quad s\in(0,t)$

$$\int_0^1 \left(\frac{1}{k}\cos(\omega s)\cos(k(x-t+s)),\ 0\right)\cdot(1,1)\,ds =$$

$$= \int \frac{1}{2}\left(\cos(\alpha+\beta) + \cos(\alpha-\beta)\right) =$$

$$= \int_0^1 \frac{1}{2}\left(\cos(s(\omega+k)+x-t) + \cos(s(\omega-k)-x+t)\right) ds$$

...

$$u_{tt} - u_{xx} = 0$$

$$u(0,x) = u_t(0,x) = \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$$

d'Alembertova formula

za velikost

$$u(x,y) = 0 + \frac{1}{2}\int_{-1}^{1} 1 \, d\xi + 0 = 1$$

$$\lim_{t \to \infty} = 1$$

$$u(x,t) = \frac{f(x-t) + f(x+t)}{2}$$

$$\sup = ?$$

$$u(x,t) \leq \frac{1}{2}(1+1) + \frac{1}{2} \cdot 2 + 0 = 2$$

$$x + t = m < 1 \qquad x > 0 \Rightarrow t < 1$$

$$u(x,t) = \frac{1}{2}(1+1) + \frac{1}{2}\int_{x-t}^{x+t} g(\xi)\, d\xi$$

$$x = 0 \quad t = m < 1$$

$$u(x,t) = 1 + \frac{1}{2}\int_{-m}^{m} g(\xi)\, d\xi = 1 + m$$

$$\lim_{m \to 1} u(x,t) = 1 \quad \Longrightarrow \quad \sup = 2$$

$$u_{tt} - u_{xx} = 0$$

$$g(x) = 0 \qquad \forall x \in [0, \infty]$$

$$u(t,0) = h(t)$$

$$u(t,0) = h(t)$$

$$u(x,t) = \frac{f(x-t) + f(x+t)}{2}$$

$$u(t,0) = \frac{1}{2}\left(f(t) + f(-t)\right) = h(t)$$

??